

A segment emanating from A with endpoint B will be denoted by $\overline{AB}$.

The important property of directed segments is that the ratio or the product of two directed segments, which are part of the same line, is assigned a sign. The sign is positive if the directed segments have the same orientation and negative otherwise. By the same logic we have

$$\overline{AB} = -\overline{BA}.$$

Now we may restate the three important theorems from this chapter in a more general way.

Theorem 1.49 (Power of a Point). *Let a line through P intersect the circle ω at two distinct points A and B . Then*

$$p(P, \omega) = \overline{PA} \cdot \overline{PB}.$$

Theorem 1.50 (Ceva's Theorem). *Let ABC be a triangle and let P, Q, R be points on the lines BC, CA, AB , respectively. Then AP, BQ, CR are concurrent or mutually parallel if and only if*

$$\frac{\overline{BP}}{\overline{PC}} \cdot \frac{\overline{CQ}}{\overline{QA}} \cdot \frac{\overline{AR}}{\overline{RB}} = 1.$$

Theorem 1.51 (Menelaus' Theorem). *Let ABC be a triangle, and let P, Q, R be points on the lines BC, CA, AB , respectively. Then the points P, Q, R are collinear if and only if*

$$\frac{\overline{BP}}{\overline{PC}} \cdot \frac{\overline{CQ}}{\overline{QA}} \cdot \frac{\overline{AR}}{\overline{RB}} = -1.$$

All of them can be proved more or less by copying proofs of their undirected versions. We leave the details to the reader.

Few Notes on Geometric Inequalities

Geometric inequalities form a wide subfield at the border of geometry and algebra. Profound exploration of the area is beyond the scope of this book so we pick and briefly discuss only the inequalities which are most significant or remarkable.

Triangle inequality

There is no doubt that the most important geometric inequality is the renowned triangle inequality.

Theorem 1.52 (Triangle inequality). *Let ABC be a triangle. Then*

$$AB + BC > AC, \quad BC + CA > BA, \quad \text{and} \quad CA + AB > CB.$$

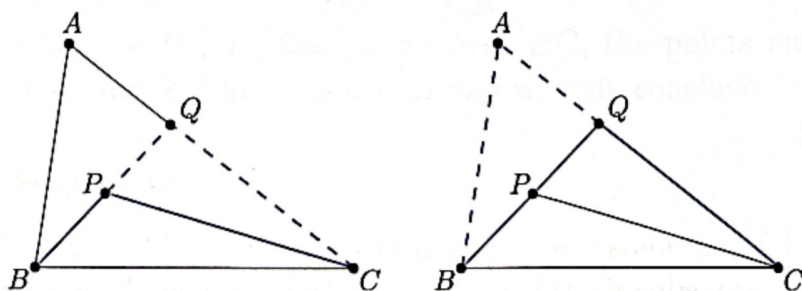
As obvious as triangle inequality may sound, it produces notable results when cleverly applied.

Example 1.25. *Let ABC be a triangle and P a point in its interior. Prove that*

$$PA + PB + PC < AB + BC + CA.$$

Proof. It seems very plausible that $BP + PC < BA + AC$. Indeed, extending BP to meet AC for the second time at Q , the triangle inequalities in triangles PCQ and ABQ yield

$$BP + PC < BQ + QC < BA + AC.$$



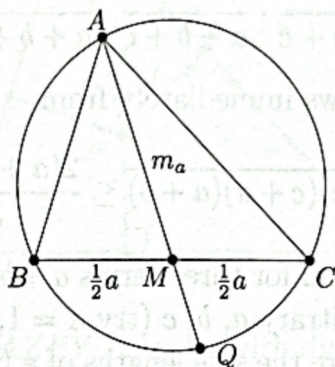
Likewise, we obtain $CP + PA < CB + BA$ and $AP + PB < AC + CB$. Summing these three inequalities and dividing by two we get the result. $\square$

Inequalities with algebraic background

The most common strategy when dealing with inequalities involving elements of a triangle is to rewrite everything in terms of independent variables.

Example 1.26. Let M, N, P be the midpoints of the sides BC, CA, AB of a triangle ABC and denote by Q, R, S the second intersections of the lines AM, BN, CP with its circumcircle ω . Prove that

$$\frac{AM}{MQ} + \frac{BN}{NR} + \frac{CP}{PS} \geq 9.$$



Proof. In order to approach the length MQ we recall Power of a Point and rewrite $MQ = (MB \cdot MC)/MA$. Keeping the median formula

$$AM^2 = \frac{1}{2}(b^2 + c^2) - \frac{1}{4}a^2$$

(see Corollary 1.24) in mind, we have just expressed everything in terms of the side lengths a, b, c . The rest is straightforward. Indeed, we may write

$$\frac{AM}{MQ} = \frac{AM^2}{MB \cdot MC} = \frac{\frac{1}{2}(b^2 + c^2) - \frac{1}{4}a^2}{(\frac{1}{2}a)^2} = \frac{2(b^2 + c^2)}{a^2} - 1$$

and similarly for other fractions. Hence it suffices to prove the inequality

$$\frac{b^2}{a^2} + \frac{c^2}{a^2} + \frac{c^2}{b^2} + \frac{a^2}{b^2} + \frac{a^2}{c^2} + \frac{b^2}{c^2} \geq 6$$

which is clearly true as it is the sum of three inequalities of the form $x + 1/x \geq 2$ for $x > 0$. $\square$

We already know that if a, b, c are the side lengths of a triangle then there exist positive numbers x, y, z such that

$$a = y + z, \quad b = x + z, \quad c = x + y$$

(these are precisely the x, y, z used in Proposition 1.26). The advantage of using x, y, z instead of a, b, c is that the former are independent positive real numbers whereas the latter have to satisfy triangle inequalities.

Example 1.27 (IMO 1991). *Prove for each triangle ABC the inequality*

$$\frac{1}{4} < \frac{IA \cdot IB \cdot IC}{l_A l_B l_C} \leq \frac{8}{27},$$

where I is the incenter and l_A, l_B, l_C are the lengths of the angle bisectors of triangle ABC .

Proof. Recalling we know the ratio in which the incenter I divides the angle bisector (see Corollary 1.28), the inequality rewrites as

$$\frac{1}{4} < \frac{b+c}{a+b+c} \cdot \frac{c+a}{a+b+c} \cdot \frac{a+b}{a+b+c} \leq \frac{8}{27}.$$

The second inequality follows immediately from

$$\sqrt[3]{(b+c)(c+a)(a+b)} \leq \frac{2(a+b+c)}{3},$$

which is just AM-GM applied for three terms $a+b, b+c, c+a$. The first one is however not true for arbitrary a, b, c (try $a=1, b=1, c=10$) so we have to use the fact that a, b, c are the side lengths of a triangle. The crucial step is to perform the x, y, z substitution which reduces the whole problem into some boring algebra. Denoting $s = x + y + z = \frac{1}{2}(a + b + c)$ it suffices to prove

$$2s^3 < (s+x)(s+y)(s+z),$$

which is true since the right-hand side expands into

$$s^3 + s^2 \underbrace{(x+y+z)}_{=s} + s(xy+yz+zx) + xyz > 2s^3. \quad \square$$

Erdős-Mordell inequality

In the very end of this section we present a famous inequality proposed by Paul Erdős¹⁵ and first proved by L. J. Mordell¹⁶. Despite its simple statement, the inequality is far from easy to prove (convince yourself!).

Theorem 1.53 (Erdős-Mordell inequality). *Let ABC be a triangle and P a point in its interior. Denote by X, Y , and Z the feet of perpendiculars dropped from P onto BC, CA , and AB , respectively. Then*

$$PA + PB + PC \geq 2(PX + PY + PZ).$$

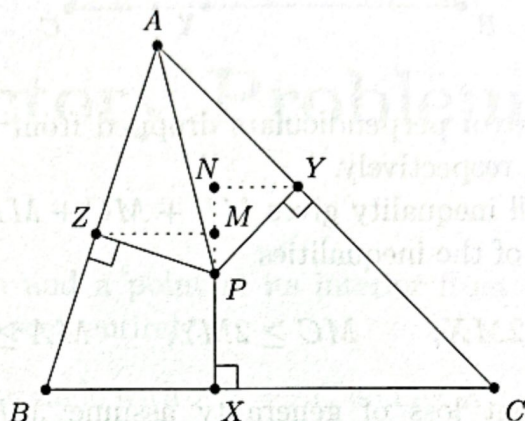
¹⁵Paul Erdős (1913–1996) was a Hungarian mathematician. He co-authored over 1500 articles and was perhaps one of the brightest minds of the 20th century.

¹⁶Louis Joel Mordell (1888–1972) was a British mathematician, known for pioneering research in number theory.

Proof. The key ingredient of the proof is the inequality

$$PA \sin \angle A \geq PY \sin \angle C + PZ \sin \angle B$$

which in fact states that the length of YZ is greater than or equal to its projection onto BC , the latter being equal to the sum of the lengths of the projections of PY and PZ .



To prove it, note that $AZPY$ is cyclic with diameter AP so the Extended Law of Sines gives $YZ = PA \sin \angle A$. Next, denote the feet of perpendiculars dropped from Z, Y onto PX by M, N , respectively. Since $BZPX$ is cyclic, we have $\angle MPZ = \angle B$ and $ZM = PZ \sin \angle B$. Likewise, $YN = PY \sin \angle C$, so the inequality is indeed equivalent to $YZ \geq YN + MZ$.

To finish the proof of the theorem, let us write the analogous two inequalities for XY and XZ . We obtain

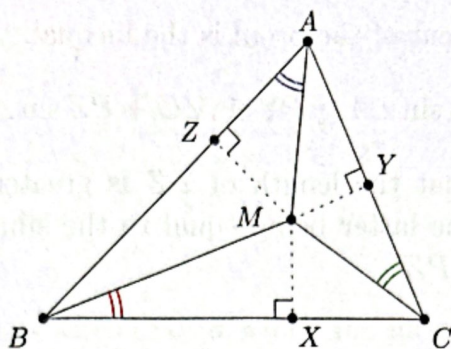
$$\begin{aligned} PA &\geq PY \cdot \frac{\sin \angle C}{\sin \angle A} + PZ \cdot \frac{\sin \angle B}{\sin \angle A}, \\ PB &\geq PZ \cdot \frac{\sin \angle A}{\sin \angle B} + PX \cdot \frac{\sin \angle C}{\sin \angle B}, \\ PC &\geq PX \cdot \frac{\sin \angle B}{\sin \angle C} + PY \cdot \frac{\sin \angle A}{\sin \angle C}. \end{aligned}$$

It remains to add these three inequalities and conclude by recalling that for positive x we have $x + 1/x \geq 2$.

The equality occurs if $YZ \parallel BC$, $ZX \parallel AC$, $XY \parallel AB$, and $\sin \angle A = \sin \angle B = \sin \angle C$, i.e. if triangle ABC is equilateral and P is its center. $\square$

Although the power of the Erdős-Mordell inequality might not be apparent at the first glance, the following example demonstrates it entirely.

Example 1.28 (IMO 1991). *Let ABC be a triangle and M a point in its interior. Show that at least one of the angles $\angle MAB$, $\angle MBC$, and $\angle MCA$ is less than or equal to 30° .*



Proof. Denote the feet of perpendiculars dropped from M to the sides BC , CA , AB by X , Y , Z , respectively.

The Erdős-Mordell inequality gives $MB + MC + MA \geq 2(MX + MY + MZ)$, so at least one of the inequalities

$$MB \geq 2MX, \quad MC \geq 2MY, \quad MA \geq 2MZ$$

has to hold. Without loss of generality assume $MB \geq 2MX$. Then $\sin MBC = MX/MB \leq \frac{1}{2}$ and thus $\angle MBC \leq 30^\circ$. $\square$

Chapter 2

Introductory Problems

1. Find a polygon and a point in its interior from which no side of the polygon can be seen entirely.
2. Let ABC be a triangle with $AB = AC$ and let K and M be points inside the side AB and L a point inside the side AC such that $BC = CM = ML = LK = KA$. Find $\angle A$.
3. Let $ABCD$ be a rectangle. Find the locus of points X such that $AX + BX = CX + DX$.
4. Let $a < b < c$ be the sides of a triangle ABC . Prove that $h_b < b$, where h_b is the B -altitude of triangle ABC .
5. On square $ABCD$, point E lies inside the side AD and point F lies inside the side BC , so that $BE = EF = FD = 30$. Find the area of square $ABCD$.
6. Let ABC be a triangle with $AB = AC$. Isosceles triangles ABM and ACN with bases AB and AC are erected outside triangle ABC . Prove that the altitudes (possibly extended) $MP \perp AB$, $NQ \perp AC$ and $AA_0 \perp BC$ are concurrent.
7. Squares $ABED$, $BCGF$, $CAIH$ are erected externally from the sides of triangle ABC . Show that triangles AID , BEF , and CGH have equal area.
8. Rhombus $ABCD$ has side length 2 and $\angle B = 120^\circ$. Region $\mathcal{R}$ consists of all points inside the rhombus that are closer to vertex B than any of the other three vertices. What is the area of $\mathcal{R}$?
9. Varignon¹ parallelogram

¹Pierre Varignon (1654–1722) was a French mathematician.

Let $ABCD$ be a quadrilateral and denote by K, L, M, N the midpoints of the sides AB, BC, CD, DA , respectively.

- (a) Prove that $KLMN$ is a parallelogram.
 - (b) Let P, Q be the midpoints of the diagonals AC, BD , respectively. Prove that $PLQN$ and $PKQM$ are also parallelograms, moreover with the same center.
10. A bus departs from the station S and rides along straight (infinite) road ℓ . Determine the locus of points in the plane from which you can catch the bus if you start running at the time of the departure and you are as fast as the bus.
 11. Point P is given inside a circle ω distinct from its center O . Determine the locus of the midpoints of the chords of ω passing through P .
 12. Let $ABCD$ be a quadrilateral inscribed in circle ω and let M_a, M_b, M_c, M_d be the midpoints of the arcs AB, BC, CD, DA not containing points C, D, A , and B , respectively. Prove that $M_a M_c \perp M_b M_d$.
 13. In rectangle $ABCD$, $AB = 9$ and $BC = 8$. Points E and F lie inside rectangle $ABCD$ so that $EF \parallel AB$, $BE \parallel DF$, $BE = 4$, $DF = 6$, and E is closer to BC than F . Find EF .
 14. Distinct points A, B, C lie on a line in this order. Circle ω_1 of radius R passing through A and B intersects circle ω_2 of the same radius R and passing through B and C for the second time at X . Find the locus of X as R varies.
 15. Let ABC be a triangle. Equilateral triangles BCD, CAE, ABF are erected outwards from its sides. Show that the circumcircles of these equilateral triangles and the lines AD, BE, CF pass through one point.
 16. Triangle ABC has right angle at B , and contains a point P for which $PA = 10$, $PB = 6$, and $\angle APB = \angle BPC = \angle CPA$. Find PC .
 17. Points P, Q are given on the sides AB, AD , respectively, of a parallelogram $ABCD$ ($AB > AD$) such that $AP = AQ = x$. Prove that as x varies, the circumcircles of the triangles PQC pass through another fixed point (other than C).
 18. In triangle ABC , medians BB_1 and CC_1 are perpendicular. Given that $AC = 19$ and $AB = 22$, find BC .

19. Let ABC be a right triangle with right angle by C and with $CA = 8$, $CB = 6$. Semicircle with diameter CX where $X \in AC$ touches side AB . Find its radius.
20. Four consecutive sides of an equiangular hexagon have lengths 1, 7, 4, and 2. Find the lengths of the remaining two sides.
21. Let ABC be a triangle with $\angle A = 60^\circ$ and denote its incenter by I . Lines BI , CI intersect the opposite sides at E , F , respectively. Prove that $IE = IF$.
22. In quadrilateral $ABCD$ let $BC = 8$, $CD = 12$, $AD = 10$, and $\angle A = \angle B = 60^\circ$. Find the distance AB .
23. Let $ABCD$ be a convex quadrilateral. Find point X for which the sum of distances to its vertices is minimal.
24. Circles ω_1 , ω_2 intersect at points A and B . An arbitrary line passing through B intersects ω_1 for the second time at K (outside ω_2) and ω_2 at L (outside ω_1).
 - (a) Prove that all possible triangles AKL are similar to each other.
 - (b) Let the tangents at points K and L to the respective circles intersect at P . Prove that $KPLA$ is cyclic.
25. Let $ABCD$ be a quadrilateral with $AB \parallel CD$. If $\angle ADB + \angle DBC = 180^\circ$, prove that

$$\frac{AB}{CD} = \frac{AD}{BC}.$$
26. On side BC of triangle ABC an arbitrary point D is selected. The tangent in D to the circumcircle of triangle ABD meets AC at point B_1 . Point C_1 is defined analogously. Prove that $B_1C_1 \parallel BC$.
27. Conway's² circle.
 Let ABC be a triangle and denote by A_1 , A_2 the points on the rays opposite to AB , AC , respectively, satisfying $AA_1 = AA_2 = BC$. Define points B_1 , B_2 , C_1 , C_2 analogously. Prove that points A_1 , A_2 , B_1 , B_2 , C_1 , C_2 lie on a single circle.
28. Let ABC be an acute triangle. Prove that $h_a > \frac{1}{2}(b + c - a)$, where h_a is the length of A -altitude in triangle ABC .

²John Horton Conway (1936) is a contemporary British mathematician known for many delightful discoveries both in recreational and research mathematics.

29. Let ABC be a triangle. Find the locus of points X ($X \neq A$) for which the triangles AXB and AXC have equal area.
30. Let $ABCD$ be a quadrilateral with perpendicular diagonals inscribed in a circle with radius R . Prove that

$$AB^2 + BC^2 + CD^2 + DA^2 = 8R^2.$$

31. A trapezoid $ABCD$ has AB parallel to CD . The external bisectors of $\angle A$ and $\angle D$ meet at P , and the external bisectors of $\angle B$ and $\angle C$ meet at Q . Show that PQ is half the perimeter of $ABCD$.
32. In triangle ABC , $11 \cdot AB = 20 \cdot AC$. The angle bisector of $\angle A$ intersects BC at point D , and point M is the midpoint of AD . Let P be the point of intersection of AC and BM . Find CP/PA .
33. A variable segment BC of fixed length d moves such that its endpoints remain on the fixed rays AU , AV . Prove that the circumcircles of all possible triangles ABC are all tangent to a fixed circle.
34. Let ABC be a right triangle with $\angle A = 90^\circ$ and altitude AD . Let r , s , t be the inradii of triangles ABC , ADB , and ADC , respectively. Show that $r + s + t = AD$.
35. In triangle ABC , $BC = 125$, $CA = 120$, and $AB = 117$. The angle bisector of angle B intersects CA at point K , and the angle bisector of angle C intersects AB at point L . Let M and N be the feet of the perpendiculars from A to CL and BK , respectively. Find MN .
36. Let $ABCD$ and $AB'C'D'$ be parallelograms such that B' lies on the segment BC and D lies on the segment $C'D'$. Show that their areas are equal.
37. In triangle ABC there is a point F on the side AB such that $\angle FAC = \angle FCB$ and $AF = BC$. Further, BE is the internal angle bisector of $\angle B$ with $E \in AC$. Show that $EF \parallel BC$.
38. Let I be the incenter of triangle ABC . Prove that

$$\frac{AI^2}{bc} + \frac{BI^2}{ca} + \frac{CI^2}{ab} = 1.$$

39. In parallelogram $ABCD$ with $\angle BAD > 90^\circ$, show that the circle passing through the projections of C onto AB , BD , and DA , respectively, passes through the center of the parallelogram.

40. Let $ABCD$ be a cyclic quadrilateral. Let P be the point on the ray AD such that $AP = BC$ and let Q be the point on the ray AB such that $AQ = CD$. Prove that the line AC cuts PQ at its midpoint.
41. Let $ABCDE$ be a convex pentagon such that $AB + CD = BC + DE$ and a circle ω with center O on the side AE is tangent to the sides AB , BC , CD and DE at points P , Q , R and S , respectively. Prove that the lines PS and AE are parallel.
42. Let P be a point inside acute-angled triangle ABC with $\angle BPC = 180 - \angle A$. Denote by A_1 , B_1 , C_1 its reflections over the sides BC , CA , AB , respectively. Prove that the points A , A_1 , B_1 , C_1 are concyclic.
43. Triangle KLM lies inside triangle ABC so that points K , L , M lie on the segments CL , AM , BK , respectively. Prove that the circumcircles of the triangles ABM , BCK , CAL pass through a common point.
44. Let the pentagon $ABCDE$ inscribed in circle ω satisfy $BA = BC$. The line joining $P = BE \cap AD$ and $Q = CE \cap BD$ intersects ω at points X , Y . Prove that $BX = BY$.
45. In the convex pentagon $ABCDE$ all interior angles have the same measure. Prove that the perpendicular bisector of segment EA , the perpendicular bisector of segment BC and the angle bisector of $\angle CDE$ intersect at one point.
46. Let ω_1 , ω_2 be two circles. One of their common external tangents is tangent to ω_1 at A , the second one is tangent to ω_2 at D . Line AD intersects the circles ω_1 , ω_2 for the second time at B , C , respectively. Prove that $AB = CD$.
47. Triangle ABC has $AB = 13$, $BC = 14$, and $CA = 15$. The points D , E , and F are the midpoints of BC , CA , and AB , respectively. Let $X \neq D$ be the intersection of the circumcircles of triangles BDF and CDE . What is $XA + XB + XC$?
48. Let $ABCD$ be a quadrilateral with segments BC and AD equal and AB not parallel to CD . Denote by M , N the midpoints of BC and AD , respectively. Prove that the perpendicular bisectors of AB , MN , and CD pass through a common point.
49. Carnot's³ Theorem.

³Lazare Nicolas Marguerite Carnot (1753–1823) was an amateur mathematician and French minister of war during the French revolutionary wars.

Let X , Y , and Z lie on the sides BC , CA , AB , respectively, of a triangle ABC . Show that the perpendiculars from X , Y , Z to the respective triangle sides meet at one point if and only if

$$BX^2 + CY^2 + AZ^2 = CX^2 + AY^2 + BZ^2.$$

50. In a given pentagon $ABCDE$, triangles ABC , BCD , CDE , DEA and EAB all have the same area. The lines AC and AD intersect BE at points M and N . Prove that $BM = EN$.
51. Let ABC be a non-right triangle with orthocenter H and let M , N be points on its sides AB and AC . Prove that the common chord of circles with diameters CM and BN passes through H .
52. Let fixed points A , Z , B lie on a line ℓ in this order such that $ZA \neq ZB$. A variable point $X \notin \ell$ and a variable point Y on the segment XZ are chosen. Let $D = BY \cap AX$ and $E = AY \cap BX$. Prove that all lines DE pass through a fixed point.
53. Let ω_1 and ω_2 be two circles centered at distinct points O_1 and O_2 and with radii r_1 , r_2 , respectively.
 - (a) Find the locus of points X for which $p(X, \omega_1) - p(X, \omega_2)$ is constant.
 - (b) Find the locus of points X for which $p(X, \omega_1) + p(X, \omega_2)$ is constant.

Chapter 3

Advanced Problems

1. On the sides AB and AD of the rhombus $ABCD$ consider the points E and F such that $AE = DF$. Let $BC \cap DE = P$ and $CD \cap BF = Q$. Prove that points P , A , and Q are collinear.
2. Let $ABCD$ be a parallelogram such that the triangle ABD is acute and has orthocenter H . The line through H parallel to AB cuts AD and BC at Q and P , respectively, while the line through H parallel to BC cuts AB and CD at R and S , respectively. Prove that the points P , Q , R , S lie on the same circle.
3. Let ABC be an acute-angled triangle. Let D and E be points on the sides AB and AC such that B , C , D , and E lie on the same circle. Further, suppose the circle through D , E , and A intersects the side BC in two points X and Y . Show that the midpoint of XY is the foot of the altitude from A to BC .
4. Point B lies on a line which is tangent to circle ω at point A . The line segment AB is rotated about the center of the circle by some angle to form segment $A'B'$. Prove that the line AA' bisects the segment BB' .
5. Let ω_1 and ω_2 be concentric circles, with ω_2 in the interior of ω_1 . From a point A on ω_1 draw the tangent AB to ω_2 ($B \in \omega_2$). Let C be the second point of intersection of AB and ω_1 , and let D be the midpoint of AB . A line passing through A intersects ω_2 at E and F in such a way that the perpendicular bisectors of DE and CF intersect at a point M on AB . Find the ratio AM/MC .
6. Let M be a point inside triangle ABC such that

$$AM \cdot BC + BM \cdot AC + CM \cdot AB = 4[ABC].$$

Show that M is the orthocenter of triangle ABC .

7. Let ABC be a triangle. Prove that lines joining midpoints of the sides with midpoints of the corresponding altitudes pass through a single point.
8. Let $ABCD$ be a convex quadrilateral such that $\angle ADB = \angle BDC$. Suppose that a point E on the side AD satisfies the equality

$$AE \cdot ED + BE^2 = CD \cdot AE.$$

Show that $\angle EBA = \angle DCB$.

9. Let ABC be a triangle with $\angle A = 90^\circ$. Denote its incenter by I and let $D = BI \cap AC$ and $E = CI \cap AB$. Determine whether or not it is possible for segments AB, AC, BI, ID, CI, IE to all have integer lengths.
10. Let A and B be two fixed points inside of the fixed circle ω symmetric with respect to its center O . If points M and N vary on ω in the same half-plane with respect to AB , so that $AM \parallel BN$, prove that $AM \cdot BN$ is constant.
11. In a trapezoid $ABCD$, the segment connecting the midpoints M, N of the bases AB, CD , respectively, has length 4, and the diagonals have lengths $AC = 6$ and $BD = 8$. Find the area of the trapezoid.
12. In triangle ABC , let AP, BQ, CR be concurrent cevians. Let the circumcircle of triangle PQR intersect the sides BC, CA, AB for the second time at X, Y, Z , respectively. Prove that AX, BY, CZ are concurrent.
13. A quadrilateral $ABCD$ is inscribed in a circle ω . The tangent to ω at B intersects the ray DC at K , and the tangent to ω at C intersects the ray AB at M . Prove that if $BM = BA$ and $CK = CD$, then $ABCD$ is a trapezoid.
14. Let $ABCD$ be a parallelogram and M, N points on its sides AB, AD such that $\angle MCB = \angle DCN$. Let P, Q, R , and S be the midpoints of the segments AB, AD, NB , and MD , respectively. Show that P, Q, R , and S are concyclic.
15. Diagonals of non-isosceles trapezoid $ABCD$ intersect at P . Let A_1 be the second intersection of the circumcircle of triangle BCD and AP . Points B_1, C_1, D_1 are defined in a similar way. Prove that $A_1B_1C_1D_1$ is also a trapezoid.
16. Let ω be a circle with center O and radius r and A a point different from O . Find the locus of circumcenters of the triangles ABC for which BC is a diameter of ω .

17. Let $ABCD$ be quadrilateral such that

$$\angle ADB + \angle ACB = 90^\circ \quad \text{and} \quad \angle DBC + 2\angle DBA = 180^\circ.$$

Show that

$$(DB + BC)^2 = AD^2 + AC^2.$$

18. We are given a triangle ABC such that $AB = AC$. There is a point D lying on the segment BC , such that $BD < DC$. Point E is symmetrical to B with respect to AD . Prove that

$$\frac{AB}{AD} = \frac{CE}{CD - BD}.$$

19. Let P be a point on the side BC of triangle ABC . Perpendicular bisectors of the sides AB and AC meet the segment AP at points D and E , respectively. The line parallel to AB passing through D intersects the tangent to the circumcircle ω of triangle ABC through B at point M . Similarly, the line parallel to AC passing through E intersects the tangent to ω through C at point N . Prove that MN is tangent to ω .
20. In an acute triangle ABC a semicircle ω centered on the side BC is tangent to the sides AB and AC at points F and E , respectively. If X is the intersection of BE and CF , show that $AX \perp BC$.
21. Let $ABCD$ be a convex quadrilateral and X a point in its interior. Denote by ω_A the circle tangent to the sides AB and AD and passing through X . Define circles ω_B , ω_C , and ω_D similarly. Given that all these circles have equal radii, show that $ABCD$ is cyclic.
22. In triangle ABC , let AP , BQ , CR be concurrent cevians. Denote by X , Y , Z the midpoints of segments QR , RP , PQ , respectively. Prove that the lines AX , BY , CZ are concurrent.
23. Given a triangle ABC , let P and Q be points on segments AB and AC , respectively, such that $AP = AQ$. Let S and R be distinct points on segment BC such that S lies between B and R , $\angle BPS = \angle PRS$, and $\angle CQR = \angle QSR$. Prove that P , Q , R , S are concyclic.
24. Segment AT is tangent to circle ω at T . A line parallel to AT intersects ω at B , C (with $AB < AC$). Lines AB , AC intersect ω for the second time at P , Q . Prove that line PQ bisects segment AT .

25. Diagonals AC and BD of a cyclic quadrilateral $ABCD$ meet at P . Let the circumcenters of $ABCD$, ABP , BCP , CDP , and DAP be O , O_1 , O_2 , O_3 , and O_4 , respectively. Prove that OP , O_1O_3 , and O_2O_4 are concurrent.
26. Triangle ABC has $BC = 20$. The incircle of the triangle evenly trisects the median AD at points E and F . Find the area of the triangle.
27. Let P be a point inside an equilateral triangle ABC . Let the lines AP , BP , CP meet the sides BC , CA , AB at the points A_1 , B_1 , C_1 , respectively. Prove that

$$A_1B_1 \cdot B_1C_1 \cdot C_1A_1 \geq A_1B \cdot B_1C \cdot C_1A.$$

28. Let P and Q be isogonal conjugates¹ with respect to the triangle ABC . Show that the six feet of perpendiculars from P and Q to the sides of triangle ABC lie on one circle.
29. The incircle of triangle ABC is tangent to its sides BC , CA , AB at points D , E , F , respectively. The excircles of triangle ABC are tangent to the corresponding sides of triangle ABC at points T , U , V . Show that triangles DEF and TUV have the same area.
30. Let H be the orthocenter of an acute-angled triangle ABC . The circle Γ_A centered at the midpoint of BC and passing through H intersects the sideline BC at points A_1 and A_2 . Similarly, define the points B_1 , B_2 , C_1 , and C_2 . Prove that six points A_1 , A_2 , B_1 , B_2 , C_1 , and C_2 are concyclic.
31. Distinct points A , B are given in the plane. Determine the locus of points C such that in triangle ABC the length of A -altitude is the same as the length of B -median.
32. Let ABC be an acute triangle with altitudes BB_0 and CC_0 . Point P is given such that the line PB is tangent to the circumcircle of triangle PAC_0 and the line PC is tangent to the circumcircle of triangle PAB_0 . Prove that AP is perpendicular to BC .
33. Let ABC be a triangle. Point O in its interior satisfies $\angle OBA = \angle OAC$, $\angle BAO = \angle OCB$, and $\angle BOC = 90^\circ$. Find AC/OC .
34. Let $ABCD$ be a cyclic quadrilateral ($AB \neq CD$). Quadrilaterals $AKDL$ and $CMBN$ are rhombi with equal sides. Prove that points K , L , M , N lie on a single circle.

¹For explanation see Theorem 1.46.

35. Let ABC be a triangle with inradius r and let ω be a circle of radius $a < r$ inscribed in angle BAC . Tangents from B and C to ω (different from the triangle sides) intersect at point X . Show that the incircle of triangle BCX is tangent to the incircle of triangle ABC .
36. Let $ABCD$ be a cyclic quadrilateral. Let P, Q, R be the feet of the perpendiculars from D to the lines BC, CA, AB , respectively. Show that $PQ = QR$ if and only if the bisectors of $\angle ABC$ and $\angle ADC$ are concurrent with AC .
37. Let X be a point on the circumcircle of a cyclic quadrilateral $ABCD$. Denote by E, F, G , and H the projections of X onto lines AB, BC, CD, DA , respectively. Prove that

$$BE \cdot CF \cdot DG \cdot AH = AE \cdot BF \cdot CG \cdot DH.$$

38. Newton-Gauss² line.

Let $ABCD$ be a convex quadrilateral. Denote by Q the intersection of AD and BC and by R the intersection of AB and CD . Let X, Y , and Z be the midpoints of AC, BD , and QR , respectively. Prove that X, Y , and Z lie on a single line.

39. In acute triangle ABC let A_1, B_1 be the points of tangency of A -excircle with BC and B -excircle with AC , respectively. Let H_1, H_2 be the orthocenters of triangles CAA_1 and CBB_1 , respectively. Prove that H_1H_2 is perpendicular to the angle bisector of $\angle ACB$.
40. A circle ω with center O is internally tangent to two circles in its interior at points S and T which are not diametrically opposite. Suppose the two circles intersect at M and N with N closer to ST . Show that $OM \perp MN$ if and only if S, N, T are collinear.
41. Let $ABCD$ be a quadrilateral with an inscribed circle ω and let the points of tangency of the incircle with sides AB, BC, CD, DA be K, L, M, N , respectively. Prove that the lines AC, BD, KM , and LN are concurrent.

42. Orthologic triangles.

Let ABC and $A'B'C'$ be two triangles in plane. Show that the perpendiculars from A' to BC , from B' to CA and from C' to AB (denote their feet by X, Y , and Z , respectively) are concurrent if and only if the perpendiculars from A to $B'C'$, from B to $C'A'$, and from C to $A'B'$ are concurrent.

²Johann Carl Friedrich Gauss (1777–1855) was a German mathematician and physicist.

43. Let ABC be a triangle with medians m_a, m_b, m_c and circumradius R . Prove that

$$\frac{b^2 + c^2}{m_a} + \frac{c^2 + a^2}{m_b} + \frac{a^2 + b^2}{m_c} \leq 12R.$$

44. Show that in acute triangle ABC we have $r + R \leq h$, where r, R , and h are the inradius, circumradius and the longest altitude, respectively.

45. Let P be a point in the plane of triangle ABC , and ℓ a line passing through P . Let A', B', C' be the points where the reflections of lines PA, PB, PC with respect to ℓ intersect lines BC, AC, AB respectively. Prove that A', B', C' are collinear.

46. Let BC be the longest side of a scalene triangle ABC . Point K on the ray CA satisfies $KC = BC$. Similarly, point L on the ray BA satisfies $BL = BC$. Prove that KL is perpendicular to OI where O, I denote the circumcenter and the incenter of triangle ABC , respectively.

47. Let D be an arbitrary point on the side BC of a given triangle ABC and let E be the intersection of AD and the second external common tangent of the incircles of triangles ABD and ACD . As D assumes all positions between B and C , prove that the point E traces an arc of a circle.

48. Let ABC be a triangle with circumcenter O . The points P and Q are interior points of the sides CA and AB , respectively. Let K, L and M be the midpoints of the segments BP, CQ and PQ , respectively, and let Γ be the circle passing through K, L , and M . Suppose that the line PQ is tangent to the circle Γ . Prove that $OP = OQ$.

49. Let ABC be a non-right triangle. A circle ω passing through B and C intersects the sides AB and AC again at C' and B' , respectively. Prove that BB', CC' and HH' are concurrent, where H and H' are the orthocenters of triangles ABC and $AB'C'$, respectively.

50. Let P be a point in the interior of triangle ABC with circumradius R . Prove that

$$\frac{AP}{a^2} + \frac{BP}{b^2} + \frac{CP}{c^2} \geq \frac{1}{R}.$$

51. Let $AXYZB$ be a convex pentagon inscribed in a semicircle of diameter AB . Denote by P, Q, R, S the feet of the perpendiculars from Y onto lines AX, BX, AZ, BZ , respectively. Prove that the acute angle formed by lines PQ and RS is half the size of $\angle ZOX$, where O is the midpoint of the segment AB .

52. Let PAB and PCD be triangles such that $PA = PB$, $PC = PD$, and triads of points P, A, C and P, D are both collinear in this order. A circle ω_1 passing through A and C intersects a circle ω_2 passing through B and D at distinct points X, Y . Prove that the circumcenter of the triangle PXY is the midpoint of the segment formed by the centers O_1, O_2 of ω_1, ω_2 , respectively.
53. Let ABC be a triangle with $\angle BCA = 90^\circ$, and let D be the foot of the altitude from C . Let X be a point in the interior of the segment CD . Let K be the point on the segment AX such that $BK = BC$. Similarly, let L be the point on the segment BX such that $AL = AC$. Let M be the point of intersection of AL and BK .

Show that $MK = ML$.

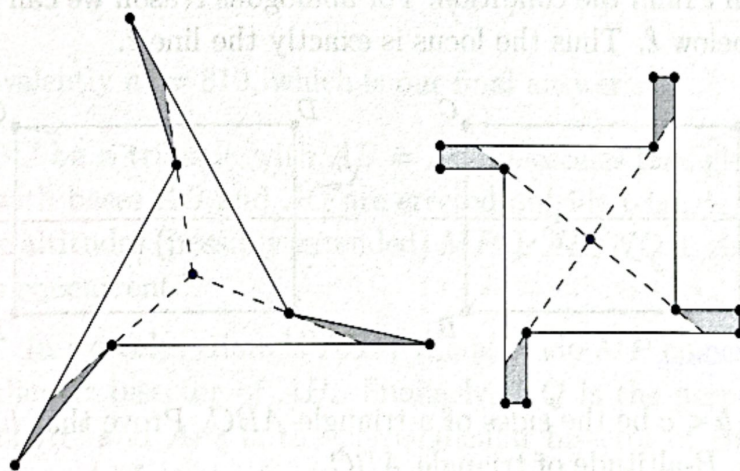


Chapter 4

Solutions to Introductory Problems

1. Find a polygon and a point in its interior from which no side of the polygon can be seen entirely.

Solution. From the many possible solutions we offer two which have similar flavor.



2. Let ABC be a triangle with $AB = AC$ and let K and M be points inside the side AB and L a point inside the side AC such that $BC = CM = ML = LK = KA$. Find $\angle A$.

Solution. Denote $\angle A$ by α . The only ingredient in this proof is that we interpret equal distances as equal angles. We have isosceles triangles BCM , CML , MLK , and LKA . Starting from triangle LKA we learn that $\angle ALK = \alpha$ and from external angle in triangle KLA we have $\angle MKL = 2\alpha$. In the same way, we obtain $\angle LMK = 2\alpha$ (triangle MLK is isosceles) and $\angle MLC = 3\alpha$ (external angle in triangle ALM). Finally,